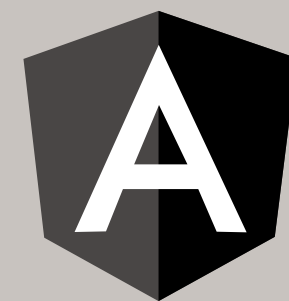


Role-based Access Control – Role Guards in Angular



Auth isn't enough — control who gets access to what.

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Why Role Guards Matter ?

- A user is logged in...
- But are they an *admin*?

Authentication $\neq$ Authorization

- Role Guards check if a user has permission to access a route.



Think: Admin-only panels, HR dashboards, billing routes.

How Role Guards Work ?

- They extend *CanActivate* or *CanLoad*, and check:
 - » Is the user logged in?
 - » Does the user have the right role?
- If ✓ allow route access.
- If ✗ redirect or show error.

Example RoleGuard Code

```
@Injectable({ providedIn: 'root' })  
export class RoleGuard implements CanActivate {  
  constructor(private auth: AuthService, private router: Router) {}  
  
  canActivate(): boolean {  
    const user = this.auth.getUser();  
    if (user && user.role === 'admin') return true;  
  
    this.router.navigate(['/unauthorized']);  
    return false;  
  }  
}
```

Customize the role logic per route.

Add It to Routes



```
{  
  path: 'admin',  
  component: AdminPanelComponent,  
  canActivate: [RoleGuard]  
}
```

Want to support **multiple roles**?

Make your guard flexible:



```
hasRole(['admin', 'manager'])
```


Dynamic Role Checking

Pass roles as data in routes:

```
{  
  path: 'settings',  
  component: SettingsComponent,  
  canActivate: [RoleGuard],  
  data: { roles: ['admin', 'editor']  
}
```

Then inside the guard:

```
const allowedRoles = route.data['roles'];
```

More scalable. Cleaner.

CanLoad for Modules Too

Block unauthorized users before the module loads:

```
canLoad(): boolean {  
  return this.auth.hasRole('admin');  
}
```



Secure lazy-loaded modules from role-based misuse.

Real-World Use Cases

Route	Allowed Roles
/admin	Admins only
/billing	Admin, Finance
/user-profile/:id	User or Admin
/moderation	Moderator, Admin

Think like a product owner.
Control everything users see.

TL;DR



Auth Guards = “Are you logged in?”



Role Guards = “Are you allowed here?”

Combine both for real access control.